

IA 642 Defensive Security

Week 05 Module 02 Lab 5.2

Live Triage on MERIDIAN-APP01: Memory, Disk, and a Recovered Sample

Digital Forensics, Incident Response & Malware Analysis

Repository: github.com/unaishhazan/ia642-week05-lab52-dfir

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Course	IA 642 Defensive Security	Week	05, Lab 5.2
Tools	WinPmem, Volatility 3, PECmd, AmcacheParser	Date	October 2026

Lab overview

This lab builds MERIDIAN-APP01 (Windows Server 2022) on `meridian-lab-net`, stages a LOLBin-typical `certutil` execution trail and an HKCU Run-key persistence entry, acquires live memory with WinPmem, triages the image with Volatility 3, parses Prefetch and Amcache, and performs a static survey of the inert `svcupdate.bin` stand-in. Evidence #1-#7, three critical-thinking answers, a taxonomy classification, and a graded hunt hypothesis follow.

Host	Role / address
MERIDIAN-APP01	Windows Server 2022 application server, 10.77.0.30/24
MERIDIAN-RED01	Analysis workstation (Volatility 3 / static triage), 10.77.0.20/24

1. Environment setup and staged artifacts

MERIDIAN-APP01 was installed from the Windows Server 2022 Evaluation ISO (Desktop Experience), attached only to `meridian-lab-net` for lab traffic, with a NAT adapter used solely to stage tools from the host. Directories `C:\Meridian` and `C:\Meridian\evidence` were created. Staging wall-clock time was recorded before artifact generation.

Section 1.2 staging

```
certutil -store My > C:\Meridian\cert_store_export.txt
certutil -hashfile C:\Windows\System32\calc.exe SHA256
certutil -hashfile C:\Windows\System32\calc.exe SHA256
Start-Sleep -Seconds 5
certutil -hashfile C:\Windows\System32\notepad.exe SHA256

reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v MeridianSyncHelper /t REG_SZ /d "powershell.exe -
NoProfile -WindowStyle Hidden -Command \"Get-Date | Out-File C:\Meridian\synclog.txt -Append\" /f
```

Staging wall-clock timestamp: 2026-10-02 20:32:15 -04:00 (captured in `C:\Meridian\staging_wallclock.txt`). Repeated `certutil` staging also logged at 2026-10-02 20:18:11 -04:00 in `pf_stage2.log`.

2. Part A: Memory acquisition and triage

Evidence #1 – WinPmem acquisition and hash

Evidence #1. WinPmem command, completion output, and SHA-256 of meridian-app01.aff4.

```
cd C:\Meridian
.\winpmem.exe -o meridian-app01.aff4 -dd
certutil -hashfile C:\Meridian\meridian-app01.aff4 SHA256
```

```
CMD (AFF4, lab-style): .\winpmem.exe -o meridian-app01.aff4 -dd
Output: C:\Meridian\meridian-app01.aff4
Size: 4418358501 bytes (snappy AFF4)
SHA256 (meridian-app01.aff4):
684ae16175c58a10084617d956b7cf5a70c802665422e497a165e349dcea6555
```

```
Also acquired raw image for Volatility 3 (go-winpmem):
CMD: .\winpmem.exe acquire --progress --verbose --nosparse meridian-app01.raw
Output: C:\Meridian\meridian-app01.raw
Size: 6979321856 bytes
SHA256 (meridian-app01.raw):
2dff14a8b7fe8f0e8426cb000690e5f06a898829fb8ccabbcc0ca03865ab494c
Volatility 3 plugins below were run against the raw image.
```

Evidence #2 – windows.pslist and windows.pstree

Evidence #2. Full Volatility 3 windows.pslist and windows.pstree output.

```
vol -f meridian-app01.aff4 windows.pslist
vol -f meridian-app01.aff4 windows.pstree
```

```
===== ev2_pslist.txt =====
Volatility 3 Framework 2.28.2

PID      PPID      ImageFileName      Offset(V)      Threads  Handles  SessionId      Wow64      CreateTime      ExitTime
4         0         System             0xcc87be881080 99        -         N/A           False      2026-10-02 22:48:42.000000 UTC  N/A
100       4         Registry           0xcc87be9ca040 4         -         N/A           False      2026-10-02 22:48:33.000000 UTC  N/A
332       4         smss.exe           0xcc87c0398040 2         -         N/A           False      2026-10-02 22:48:42.000000 UTC  N/A
428       420       csrss.exe          0xcc87c20b5080 9         -         0             False      2026-10-02 22:48:44.000000 UTC  N/A
496       488       csrss.exe          0xcc87c21f3080 10        -         1             False      2026-10-02 22:48:45.000000 UTC  N/A
504       420       wininit.exe        0xcc87c21f1140 1         -         0             False      2026-10-02 22:48:45.000000 UTC  N/A
560       488       winlogon.exe       0xcc87c2e36080 5         -         1             False      2026-10-02 22:48:45.000000 UTC  N/A
624       504       services.exe       0xcc87c2e4d080 6         -         0             False      2026-10-02 22:48:45.000000 UTC  N/A
636       504       lsass.exe          0xcc87c2e790c0 9         -         0             False      2026-10-02 22:48:45.000000 UTC  N/A
744       624       svchost.exe        0xcc87c2f3a240 9         -         0             False      2026-10-02 22:48:46.000000 UTC  N/A
776       504       fontdrvhost.ex     0xcc87c2ee4140 5         -         0             False      2026-10-02 22:48:46.000000 UTC  N/A
784       560       fontdrvhost.ex     0xcc87c2f42140 5         -         1             False      2026-10-02 22:48:46.000000 UTC  N/A
868       624       svchost.exe        0xcc87c2f512c0 9         -         0             False      2026-10-02 22:48:46.000000 UTC  N/A
924       624       svchost.exe        0xcc87c2f83240 4         -         0             False      2026-10-02 22:48:46.000000 UTC  N/A
992       560       dwm.exe            0xcc87c2feb080 17        -         1             False      2026-10-02 22:48:47.000000 UTC  N/A
424       624       svchost.exe        0xcc87c302d300 3         -         0             False      2026-10-02 22:48:47.000000 UTC  N/A
376       624       svchost.exe        0xcc87c3030300 3         -         0             False      2026-10-02 22:48:47.000000 UTC  N/A
392       624       svchost.exe        0xcc87c3032080 2         -         0             False      2026-10-02 22:48:47.000000 UTC  N/A
1060      624       svchost.exe        0xcc87c30652c0 2         -         0             False      2026-10-02 22:48:47.000000 UTC  N/A
1076      624       svchost.exe        0xcc87c3070300 7         -         0             False      2026-10-02 22:48:47.000000 UTC  N/A
1212      624       svchost.exe        0xcc87c3061080 3         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1264      624       VBoxService.ex     0xcc87c317b080 14        -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1300      624       svchost.exe        0xcc87c317a080 3         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1336      624       svchost.exe        0xcc87c3179080 6         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1356      624       svchost.exe        0xcc87c3178080 1         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1372      624       svchost.exe        0xcc87c3177080 3         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1380      624       svchost.exe        0xcc87c3176080 4         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1388      624       svchost.exe        0xcc87c3175080 4         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1408      624       svchost.exe        0xcc87c3174080 2         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1508      624       svchost.exe        0xcc87c3173080 7         -         0             False      2026-10-02 22:48:48.000000 UTC  N/A
1656      624       svchost.exe        0xcc87c3172080 2         -         0             False      2026-10-02 22:48:49.000000 UTC  N/A
1708      624       svchost.exe        0xcc87c3171080 5         -         0             False      2026-10-02 22:48:49.000000 UTC  N/A
1764      624       svchost.exe        0xcc87c3170080 9         -         0             False      2026-10-02 22:48:49.000000 UTC  N/A
1776      624       svchost.exe        0xcc87c316f080 4         -         0             False      2026-10-02 22:48:49.000000 UTC  N/A
1808      624       svchost.exe        0xcc87be92c080 3         -         0             False      2026-10-02 22:48:49.000000 UTC  N/A
```

1852	624	svchost.exe	0xcc87be938080	11	-	0	False	2026-10-02	22:48:49.000000	UTC	N/A
1928	624	svchost.exe	0xcc87be93b080	5	-	0	False	2026-10-02	22:48:49.000000	UTC	N/A
1940	624	svchost.exe	0xcc87be939080	4	-	0	False	2026-10-02	22:48:49.000000	UTC	N/A
1960	624	svchost.exe	0xcc87be93a080	3	-	0	False	2026-10-02	22:48:49.000000	UTC	N/A
1156	624	svchost.exe	0xcc87be933080	4	-	0	False	2026-10-02	22:48:49.000000	UTC	N/A
2224	624	spoolsv.exe	0xcc87be93d080	7	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2264	624	svchost.exe	0xcc87be940080	8	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2280	624	svchost.exe	0xcc87be93c080	10	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2300	624	svchost.exe	0xcc87be93e080	5	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2388	624	svchost.exe	0xcc87be941080	2	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2400	624	svchost.exe	0xcc87be942080	2	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2412	624	svchost.exe	0xcc87be943080	3	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2432	624	svchost.exe	0xcc87be944080	5	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2444	624	svchost.exe	0xcc87be932080	4	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2488	624	MsMpEng.exe	0xcc87be9460c0	25	-	0	False	2026-10-02	22:48:50.000000	UTC	N/A
2524	624	svchost.exe	0xcc87be948080	11	-	0	False	2026-10-02	22:48:51.000000	UTC	N/A
2572	624	wlms.exe	0xcc87be949080	2	-	0	False	2026-10-02	22:48:51.000000	UTC	N/A
2584	624	svchost.exe	0xcc87be94a080	10	-	0	False	2026-10-02	22:48:51.000000	UTC	N/A
2688	624	svchost.exe	0xcc87be94b080	2	-	0	False	2026-10-02	22:48:51.000000	UTC	N/A
2668	2280	AggregatordHost	0xcc87be94e080	2	-	0	False	2026-10-02	22:48:52.000000	UTC	N/A
3176	1508	MicrosoftEdgeU	0xcc87be94d080	4	-	0	True	2026-10-02	22:48:54.000000	UTC	N/A
3200	1156	sihost.exe	0xcc87c316e240	9	-	1	False	2026-10-02	22:48:54.000000	UTC	N/A
3208	624	svchost.exe	0xcc87c316c0c0	3	-	1	False	2026-10-02	22:48:54.000000	UTC	N/A
3272	624	svchost.exe	0xcc87c3955300	3	-	1	False	2026-10-02	22:48:54.000000	UTC	N/A
3352	1508	taskhostw.exe	0xcc87c399d0c0	2	-	1	False	2026-10-02	22:48:54.000000	UTC	N/A
3488	624	svchost.exe	0xcc87c39d1240	2	-	0	False	2026-10-02	22:48:54.000000	UTC	N/A
3540	624	svchost.exe	0xcc87c39d4300	3	-	0	False	2026-10-02	22:48:55.000000	UTC	N/A
3580	3540	ctfmon.exe	0xcc87c39ea280	8	-	1	False	2026-10-02	22:48:55.000000	UTC	N/A
3752	624	svchost.exe	0xcc87c3adc240	2	-	0	False	2026-10-02	22:48:55.000000	UTC	N/A
3808	560	userinit.exe	0xcc87c3adf340	0	-	1	False	2026-10-02	22:48:56.000000	UTC	
2026-10-02 22:49:20.000000 UTC											
3816	624	svchost.exe	0xcc87c3ae0080	3	-	0	False	2026-10-02	22:48:56.000000	UTC	N/A
3860	3808	explorer.exe	0xcc87c3ae5340	29	-	1	False	2026-10-02	22:48:56.000000	UTC	N/A
2888	624	NisSrv.exe	0xcc87c3b4e080	5	-	0	False	2026-10-02	22:48:57.000000	UTC	N/A
3464	744	StartMenuExper	0xcc87be936080	7	-	1	False	2026-10-02	22:48:59.000000	UTC	N/A
3996	744	TextInputHost.	0xcc87c316a080	10	-	1	False	2026-10-02	22:48:59.000000	UTC	N/A
4128	744	RuntimeBroker.	0xcc87c399b080	1	-	1	False	2026-10-02	22:49:00.000000	UTC	N/A
4268	744	SearchApp.exe	0xcc87c3d53080	16	-	1	False	2026-10-02	22:49:00.000000	UTC	N/A
4392	744	RuntimeBroker.	0xcc87c3d95300	1	-	1	False	2026-10-02	22:49:01.000000	UTC	N/A
4500	3340	ServerManager.	0xcc87c3e69380	10	-	1	False	2026-10-02	22:49:02.000000	UTC	N/A
5068	3860	VBoxTray.exe	0xcc87c4058340	12	-	1	False	2026-10-02	22:49:13.000000	UTC	N/A
3812	624	svchost.exe	0xcc87c3ce6080	22	-	0	False	2026-10-02	22:50:52.000000	UTC	N/A
4312	624	msdtc.exe	0xcc87be94c080	9	-	0	False	2026-10-02	22:50:52.000000	UTC	N/A
5036	624	svchost.exe	0xcc87c3d7f080	2	-	0	False	2026-10-02	22:50:53.000000	UTC	N/A
3956	624	svchost.exe	0xcc87c4196080	9	-	0	False	2026-10-02	22:50:53.000000	UTC	N/A
2024	624	svchost.exe	0xcc87c41ad300	4	-	1	False	2026-10-02	22:50:54.000000	UTC	N/A
2516	624	svchost.exe	0xcc87c419c2c0	7	-	0	False	2026-10-02	22:50:55.000000	UTC	N/A
1832	744	wsmprovhost.ex	0xcc87be8f4080	14	-	0	False	2026-10-02	22:56:41.000000	UTC	N/A
3468	1832	net.exe	0xcc87be911080	0	-	0	False	2026-10-02	22:56:44.000000	UTC	2026-10-02
22:56:44.000000 UTC											
4244	744	wsmprovhost.ex	0xcc87c2f39080	21	-	0	False	2026-10-02	23:07:29.000000	UTC	N/A
2164	4244	winpmem.exe	0xcc87be8da080	8	-	0	False	2026-10-02	23:07:31.000000	UTC	N/A
2852	2164	conhost.exe	0xcc87be93f080	6	-	0	False	2026-10-02	23:07:31.000000	UTC	N/A
===== ev2_pstree.txt =====											
Volatility 3 Framework 2.28.2											
PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime		
4	0	System	0xcc87be881080	99	-	N/A	False	2026-10-02 22:48:42.000000	UTC	N/A	-
* 100	4	Registry	0xcc87be9ca040	4	-	N/A	False	2026-10-02 22:48:33.000000	UTC	N/A	
* 332	4	smss.exe	0xcc87c0398040	2	-	N/A	False	2026-10-02 22:48:42.000000	UTC	N/A	
		\Device\HarddiskVolume1\Windows\System32\smss.exe						\SystemRoot\System32\smss.exe		\SystemRoot\	
428	420	csrss.exe	0xcc87c20b5080	9	-	0	False	2026-10-02 22:48:44.000000	UTC	N/A	
		\Device\HarddiskVolume1\Windows\System32\csrss.exe						%SystemRoot%\system32\csrss.exe		ObjectDirectory=\	
		Windows SharedSection=1024,20480,768 Windows=0n SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:						UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16		C:\Windows\	
		system32\csrss.exe									

```

496      488      csrss.exe      0xcc87c21f3080 10      -      1      False      2026-10-02 22:48:45.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\csrss.exe      %SystemRoot%\system32\csrss.exe ObjectDirectory=\
Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:
UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16 C:\Windows\
system32\csrss.exe
504      420      wininit.exe      0xcc87c21f1140 1      -      0      False      2026-10-02 22:48:45.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\wininit.exe      wininit.exe C:\Windows\system32\wininit.exe
* 624      504      services.exe      0xcc87c2e4d080 6      -      0      False      2026-10-02 22:48:45.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\services.exe      C:\Windows\system32\services.exe C:\
Windows\system32\services.exe
** 1408 624      svchost.exe      0xcc87c3174080 2      -      0      False      2026-10-02 22:48:48.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\System32\svchost.exe -k netsvcs -p -s
Themes C:\Windows\System32\svchost.exe
** 2432 624      svchost.exe      0xcc87be944080 5      -      0      False      2026-10-02 22:48:50.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\System32\svchost.exe -k smbssvc -s
LanmanServer C:\Windows\System32\svchost.exe
** 2688 624      svchost.exe      0xcc87be94b080 2      -      0      False      2026-10-02 22:48:51.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\system32\svchost.exe -k netsvcs -p -s
WpnService C:\Windows\system32\svchost.exe
** 1156 624      svchost.exe      0xcc87be933080 4      -      0      False      2026-10-02 22:48:49.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\system32\svchost.exe -k netsvcs -p -s
UserManager C:\Windows\system32\svchost.exe
*** 3200      1156      sihost.exe      0xcc87c316e240 9      -      1      False      2026-10-02 22:48:54.000000
UTC N/A      \Device\HarddiskVolume1\Windows\System32\sihost.exe      sihost.exe C:\Windows\system32\
sihost.exe
** 392 624      svchost.exe      0xcc87c3032080 2      -      0      False      2026-10-02 22:48:47.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\System32\svchost.exe -k
LocalSystemNetworkRestricted -p -s NcbService C:\Windows\System32\svchost.exe
** 1928 624      svchost.exe      0xcc87be93b080 5      -      0      False      2026-10-02 22:48:49.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\System32\svchost.exe -k NetworkService
-p -s LanmanWorkstation C:\Windows\System32\svchost.exe
** 3208 624      svchost.exe      0xcc87c316c0c0 3      -      1      False      2026-10-02 22:48:54.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\system32\svchost.exe -k
UnistackSvcGroup -s CDPUserSvc C:\Windows\system32\svchost.exe
** 2444 624      svchost.exe      0xcc87be932080 4      -      0      False      2026-10-02 22:48:50.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\system32\svchost.exe -k LocalService -
s W32Time C:\Windows\system32\svchost.exe
** 2572 624      wlms.exe      0xcc87be949080 2      -      0      False      2026-10-02 22:48:51.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\wlms\wlms.exe      C:\Windows\system32\wlms\wlms.exe C:\
Windows\system32\wlms\wlms.exe
** 1808 624      svchost.exe      0xcc87be92c080 3      -      0      False      2026-10-02 22:48:49.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\System32\svchost.exe -k netsvcs -p -s
ShellHWDetection C:\Windows\System32\svchost.exe
** 1300 624      svchost.exe      0xcc87c317a080 3      -      0      False      2026-10-02 22:48:48.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\system32\svchost.exe -k netsvcs -p -s
gpsvc C:\Windows\system32\svchost.exe
** 1940 624      svchost.exe      0xcc87be939080 4      -      0      False      2026-10-02 22:48:49.000000 UTC N/A
      \Device\HarddiskVolume1\Windows\System32\svchost.exe      C:\Windows\System32\svchost.exe -k LocalService -
p
... [truncated for report length] ...

```

Evidence #3 – windows.netscan and windows.malfind

Evidence #3. Full Volatility 3 windows.netscan and windows.malfind output.

```

vol -f meridian-app01.aff4 windows.netscan
vol -f meridian-app01.aff4 windows.malfind

```

```

===== ev3_netscan.txt =====
Volatility 3 Framework 2.28.2
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
0xcc87be8a5730 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 1508 svchost.exe 2026-10-02
22:48:49.000000 UTC
0xcc87be8a5890 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 1508 svchost.exe 2026-10-02
22:48:49.000000 UTC
0xcc87be8a5890 TCPv6 :: 49667 :: 0 LISTENING 1508 svchost.exe 2026-10-02
22:48:49.000000 UTC
0xcc87be8a59f0 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 2224 spoolsv.exe 2026-10-02
22:48:50.000000 UTC

```

0xcc87be8a5b50	TCPv4	0.0.0.0	49668	0.0.0.0	0	LISTENING	2224	spoolsv.exe	2026-10-02	22:48:50.000000	UTC	
0xcc87be8a5b50	TCPv6	::	49668	::	0	LISTENING	2224	spoolsv.exe	2026-10-02	22:48:50.000000	UTC	
0xcc87be8a5cb0	TCPv4	0.0.0.0	49669	0.0.0.0	0	LISTENING	624	services.exe	2026-10-02	22:48:52.000000	UTC	
0xcc87be8a5cb0	TCPv6	::	49669	::	0	LISTENING	624	services.exe	2026-10-02	22:48:52.000000	UTC	
0xcc87c206ca60	TCPv4	10.0.2.15		5985	10.0.2.2	56575	CLOSED	4	System	2026-10-02	22:49:20.000000	UTC
0xcc87c2e27310	TCPv4	10.77.0.30		139	0.0.0.0	0	LISTENING	4	System	2026-10-02	22:48:48.000000	UTC
0xcc87c2e27470	TCPv4	0.0.0.0	49665	0.0.0.0	0	LISTENING	504	wininit.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e27470	TCPv6	::	49665	::	0	LISTENING	504	wininit.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e275d0	TCPv4	0.0.0.0	49664	0.0.0.0	0	LISTENING	636	lsass.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e275d0	TCPv6	::	49664	::	0	LISTENING	636	lsass.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e27890	TCPv4	0.0.0.0	49666	0.0.0.0	0	LISTENING	1076	svchost.exe	2026-10-02	22:48:48.000000	UTC	
0xcc87c2e27b50	TCPv4	0.0.0.0	49669	0.0.0.0	0	LISTENING	624	services.exe	2026-10-02	22:48:52.000000	UTC	
0xcc87c2e27cb0	TCPv4	0.0.0.0	445	0.0.0.0	0	LISTENING	4	System	2026-10-02	22:48:52.000000	UTC	
0xcc87c2e27cb0	TCPv6	::	445	::	0	LISTENING	4	System	2026-10-02	22:48:52.000000	UTC	
0xcc87c2e28230	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	868	svchost.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e28390	TCPv4	0.0.0.0	49665	0.0.0.0	0	LISTENING	504	wininit.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e284f0	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	868	svchost.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e284f0	TCPv6	::	135	::	0	LISTENING	868	svchost.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e28650	TCPv4	10.0.2.15		139	0.0.0.0	0	LISTENING	4	System	2026-10-02	22:48:49.000000	UTC
0xcc87c2e28910	TCPv4	0.0.0.0	49664	0.0.0.0	0	LISTENING	636	lsass.exe	2026-10-02	22:48:46.000000	UTC	
0xcc87c2e28e90	TCPv4	0.0.0.0	49666	0.0.0.0	0	LISTENING	1076	svchost.exe	2026-10-02	22:48:48.000000	UTC	
0xcc87c2e28e90	TCPv6	::	49666	::	0	LISTENING	1076	svchost.exe	2026-10-02	22:48:48.000000	UTC	
0xcc87c32994e0	UDPv4	10.77.0.30		138	*	0	4	System	2026-10-02	22:48:48.000000	UTC	
0xcc87c3299670	UDPv4	10.77.0.30		137	*	0	4	System	2026-10-02	22:48:48.000000	UTC	
0xcc87c329fc00	UDPv6	fe80::c006:d141:cc22:9d2f		546	*	0		1336	svchost.exe	2026-10-02	22:48:49.000000	UTC
0xcc87c32a1820	UDPv4	10.0.2.15		138	*	0	4	System	2026-10-02	22:48:49.000000	UTC	
0xcc87c32a19b0	UDPv4	10.0.2.15		137	*	0	4	System	2026-10-02	22:48:49.000000	UTC	
0xcc87c32a43e0	UDPv6	fe80::84c3:2d82:c66:3357		546	*	0		1336	svchost.exe	2026-10-02	22:48:49.000000	UTC
0xcc87c3391820	UDPv4	0.0.0.0	5355	*	0	1764	svchost.exe	2026-10-02	22:48:49.000000	UTC		
0xcc87c3392ae0	UDPv4	0.0.0.0	5353	*	0	1764	svchost.exe	2026-10-02	22:48:49.000000	UTC		
0xcc87c33932b0	UDPv4	0.0.0.0	0	*	0	1764	svchost.exe	2026-10-02	22:48:49.000000	UTC		
0xcc87c33932b0	UDPv6	::	0	*	0	1764	svchost.exe	2026-10-02	22:48:49.000000	UTC		
0xcc87c3393440	UDPv4	0.0.0.0	5353	*	0	1764	svchost.exe	2026-10-02	22:48:49.000000	UTC		
0xcc87c3393440	UDPv6	::	5353	*	0	1764	svchost.exe	2026-10-02	22:48:49.000000	UTC		
0xcc87c33940c0	UDPv4	0.0.0.0	5355	*	0	1764	svchost.exe	2026-10-02	22:48:49.000000	UTC		

0xcc87c33940c0	UDPv6	::	5355	*	0	1764	svchost.exe	2026-10-02 22:48:49.000000	UTC
0xcc87c35d4820	UDPv4	127.0.0.1	52363	*	0	2300	svchost.exe	2026-10-02	22:48:52.000000 UTC
0xcc87c35d54a0	UDPv4	0.0.0.0 0	*	0	2444	svchost.exe	2026-10-02 22:48:51.000000	UTC	
0xcc87c35d54a0	UDPv6	::	0	*	0	2444	svchost.exe	2026-10-02 22:48:51.000000	UTC
0xcc87c35d70c0	UDPv4	0.0.0.0 0	*	0	2444	svchost.exe	2026-10-02 22:48:51.000000	UTC	
0xcc87c3613a60	TCPv4	10.0.2.15	5985	10.0.2.2	54405	ESTABLISHED	4	System	2026-10-02 23:07:30.000000 UTC
0xcc87c379cb50	TCPv4	0.0.0.0 47001	0.0.0.0 0	LISTENING	4	System	2026-10-02 22:48:53.000000	UTC	
0xcc87c379cb50	TCPv6	::	47001	::	0	LISTENING	4	System	2026-10-02 22:48:53.000000
0xcc87c379ccb0	TCPv4	0.0.0.0 5985	0.0.0.0 0	LISTENING	4	System	2026-10-02 22:48:53.000000	UTC	
0xcc87c379ccb0	TCPv6	::	5985	::	0	LISTENING	4	System	2026-10-02 22:48:53.000000
0xcc87c37dca40	UDPv4	0.0.0.0 123	*	0	2444	svchost.exe	2026-10-02 22:48:52.000000	UTC	
0xcc87c37dca40	UDPv6	::	123	*	0	2444	svchost.exe	2026-10-02 22:48:52.000000	UTC
0xcc87c37dd3a0	UDPv4	0.0.0.0 123	*	0	2444	svchost.exe	2026-10-02 22:48:52.000000	UTC	
0xcc87c37f9d70	UDPv4	0.0.0.0 51007	*	0	1764	svchost.exe	2026-10-02 22:48:55.000000	UTC	
0xcc87c37f9d70	UDPv6	::	51007	*	0	1764	svchost.exe	2026-10-02 22:48:55.000000	UTC
0xcc87c3a20460	TCPv4	10.0.2.15	5985	10.0.2.2	54404	ESTABLISHED	4	System	2026-10-02 23:07:29.000000 UTC
0xcc87c3b1a910	UDPv4	0.0.0.0 0	*	0	1264	VBxService.exe	2026-10-02 23:07:38.000000	UTC	
0xcc87c3b4da20	TCPv4	10.0.2.15	49681	48.211.71.196	443	ESTABLISHED	2488	MsMpEng.exe	2026-10-02 23:07:32.000000 UTC
0xcc87c3b699a0	TCPv4	10.0.2.15	49676	172.178.160.22	443	CLOSED	2488	MsMpEng.exe	2026-10-02 22:49:00.000000 UTC
0xcc87c3e714a0	TCPv4	10.0.2.15	5985	10.0.2.2	54408	ESTABLISHED	4	System	2026-10-02 23:07:30.000000 UTC
0xcc87c3eef260	UDPv4	0.0.0.0 0	*	0	2488	MsMpEng.exe	2026-10-02 23:07:32.000000	UTC	
0xcc87c3eef260	UDPv6	::	0	*	0	2488	MsMpEng.exe	2026-10-02 23:07:32.000000	UTC
0xcc87c3f6ab60	TCPv4	10.0.2.15	49680	48.211.71.196	443	ESTABLISHED	2488	MsMpEng.exe	2026-10-02 23:07:32.000000 UTC
0xcc87c4063410	TCPv4	10.0.2.15	5985	10.0.2.2	56705	CLOSED	4	System	2026-10-02 22:49:41.000000 UTC
0xcc87c4176a60	TCPv4	10.0.2.15	5985	10.0.2.2	56563	CLOSED	4	System	2026-10-02 22:49:18.000000 UTC
0xcc87c4232d20	UDPv4	0.0.0.0 63064	*	0	1764	svchost.exe	2026-10-02 23:07:32.000000	UTC	
0xcc87c4232d20	UDPv6	::	63064	*	0	1764	svchost.exe	2026-10-02 23:07:32.000000	UTC
===== ev3_malfind.txt =====									
Volatility 3 Framework 2.28.2									
PID	Process	Start	VPN	End	VPN	Tag	Protection	CommitCharge	PrivateMemory
Disasm								Notes	Hexdump
2488	MsMpEng.exe	0x1f321240000	0x1f32133ffff	VadS	PAGE_EXECUTE_READWRITE	256	1	N/A	
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc									
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc									
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc									
cc 08 00 42 00 00 00 05 48 8b 45 20 48 89 c2	...B....H.E H..								
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc									
42 00 00 00 00 05 48 8b 45 20 48 89 c2									
2488	MsMpEng.exe	0x1f321350000	0x1f32154ffff	VadS	PAGE_EXECUTE_READWRITE	512	1	N/A	
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc									
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc									
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc									

```

cc 09 00 38 00 09 00 01 05 ba fc ff ff ff 03 55 ...8.....U      cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc
cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc cc 09 00
38 00 09 00 01 05 ba fc ff ff ff 03 55
2488  MsMpEng.exe      0x1f321c00000 0x1f321d0cfff VadS  PAGE_EXECUTE_READWRITE 269 1 N/A
56 57 53 55 41 54 41 55 41 56 41 57 48 83 ec 28 VWSUATAUAVAWH..(
4c 8d 3c 24 48 8b e9 48 8d b1 98 38 00 00 ff e2 L.<$H..H...8....
49 8d 67 28 41 5f 41 5e 41 5d 41 5c 5d 5b 5f 5e I.g(A_A^A]A\)[_~
c3 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..... 56 57 53 55 41 54 41 55 41 56 41 57 48 83
ec 28 4c 8d 3c 24 48 8b e9 48 8d b1 98 38 00 00 ff e2 49 8d 67 28 41 5f 41 5e 41 5d 41 5c 5d 5b 5f 5e c3 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
2488  MsMpEng.exe      0x1f3215d0000 0x1f3216dcfff VadS  PAGE_EXECUTE_READWRITE 269 1 N/A
56 57 53 55 41 54 41 55 41 56 41 57 48 83 ec 28 VWSUATAUAVAWH..(
4c 8d 3c 24 48 8b e9 48 8d b1 98 38 00 00 ff e2 L.<$H..H...8....
49 8d 67 28 41 5f 41 5e 41 5d 41 5c 5d 5b 5f 5e I.g(A_A^A]A\)[_~
c3 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..... 56 57 53 55 41 54 41 55 41 56 41 57 48 83
ec 28 4c 8d 3c 24 48 8b e9 48 8d b1 98 38 00 00 ff e2 49 8d 67 28 41 5f 41 5e 41 5d 41 5c 5d 5b 5f 5e c3 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
4500  ServerManager. 0x1c5e3770000 0x1c5e377cfff VadS  PAGE_EXECUTE_READWRITE 1 1 N/A
00 00 00 00 00 00 00 00 40 44 3e c9 c5 01 00 00 .....@D>.....
40 44 3e c9 c5 01 00 00 00 00 00 3e c9 c5 01 00 00 @D>.....>.....
e0 0d 77 e3 c5 01 00 00 00 10 77 e3 c5 01 00 00 ..w.....w.....
00 d0 77 e3 c5 01 00 00 03 00 00 00 00 00 00 00 00 ..w..... 00 00 00 00 00 00 00 00 40 44 3e c9 c5 01
00 00 40 44 3e c9 c5 01 00 00 00 00 3e c9 c5 01 00 00 e0 0d 77 e3 c5 01 00 00 00 10 77 e3 c5 01 00 00 00 d0 77
e3 c5 01 00 00 03 00 00 00 00 00 00 00 00 00
4500  ServerManager. 0x7ff440d20000 0x7ff440d2ffff VadS  PAGE_EXECUTE_READWRITE 1 1 N/A
0d 00 00 00 00 02 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ..... 0d 00 00 00 00 02 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
4500  ServerManager. 0x7ff440d30000 0x7ff440dcffff VadS  PAGE_EXECUTE_READWRITE 2 1 N/A
00 1e 7c 14 f3 01 00 00 80 1c 7c 14 f3 01 00 00 ..|.....|.....
46 85 69 00 00 00 00 00 00 00 00 00 00 00 00 08 00 F.i.....
c0 94 2d 16 f3 01 00 00 c0 fe 7b 14 f3 01 00 00 ..-.....{.....
2f 89 ba 14 f3 01 00 00 6a 06 00 00 00 00 00 00 00 /.....j..... 00 1e 7c 14 f3 01 00 00 80 1c 7c 14 f3 01
00 00 46 85 69 00 00 00 00 00 00 00 00 00 00 00 08 00 c0 94 2d 16 f3 01 00 00 c0 fe 7b 14 f3 01 00 00 2f 89 ba
14 f3 01 00 00 6a 06 00 00 00 00 00 00 00
1832  wsmprovhost.ex 0x7df4f6bb0000 0x7df4f6bbffff VadS  PAGE_EXECUTE_READWRITE 1 1 N/A
00 00 00 00 00 00 00 00 78 0d 00 00 00 00 00 00 .....x.....
0c 00 00 00 49 c7 c2 00 00 00 00 48 b8 50 4a ff ....I.....H.PJ.
bf ff 7f 00 00 ff e0 49 c7 c2 01 00 00 00 48 b8 .....I.....H.
50 4a ff bf ff 7f 00 00 ff e0 49 c7 c2 02 00 00 PJ.....I..... 00 00 00 00 00 00 00 00 78 0d 00 00 00 00
00 00 0c 00 00 00 49 c7 c2 00 00 00 00 48 b8 50 4a ff bf ff 7f 00 00 ff e0 49 c7 c2 01 00 00 00 48 b8 50 4a ff
bf ff 7f 00 00 ff e0 49 c7 c2 02 00 00
1832  wsmprovhost.ex 0x2aee02d0000 0x2aee02dcfff VadS  PAGE_EXECUTE_READWRITE 1 1 N/A
00 00 00 00 00 00 00 00 40 75 82 f8 ae 02 00 00 .....@u.....
40 75 82 f8 ae 02 00 00 00 00 00 82 f8 ae 02 00 00 @u.....
e0 0d 2d e0 ae 02 00 00 00 10 2d e0 ae 02 00 00 ..-.....-.....
00 d0 2d e0 ae 02 00 00 01 00 00 00 00 00 00 00 00 ..-..... 00 00 00 00 00 00 00 00 40 75 82 f8 ae 02
00 00 40 75 82 f8 ae 02 00 00 00 00 82 f8 ae 02 00 00 e0 0d 2d e0 ae 02 00 00 00 10 2d e0 ae 02 00 00 00 d0 2d
e0 ae 02 00 00 01 00 00 00 00 00 00
1832  wsmprovhost.ex 0x7df4f6bc0000 0x7df4f6c5ffff VadS  PAGE_EXECUTE_READWRITE 2 1 N/A
d8 ff ff ff ff ff ff 08 00 00 00 00 00 00 00 .....
01 00 00 00 00 00 00 00 00 02 0e 03 38 00 00 00 .....8...
68 01 d7 07 0c 00 00 00 d8 5d 95 be ff 7f 00 00 h.....].....
00 10 93 be ff 7f 00 00 08 4a ac be ff 7f 00 00 .....J..... d8 ff ff ff ff ff ff 08 00 00 00 00 00
00 00 01 00 00 00 00 00 00 00 00 02 0e 03 38 00 00 00 68 01 d7 07 0c 00 00 00 d8 5d 95 be ff 7f 00 00 00 10 93
be ff 7f 00 00 08 4a ac be ff 7f 00 00
4244  wsmprovhost.ex 0x7df472aa0000 0x7df472aaffff VadS  PAGE_EXECUTE_READWRITE 1 1 N/A
00 00 00 00 00 00 00 00 78 0d 00 00 00 00 00 00 .....x.....
0c 00 00 00 49 c7 c2 00 00 00 00 48 b8 50 4a ff ....I.
... [truncated for report length] ...

```

Critical thinking #1

In the acquisition-time `pstree/cmdline` output, `certutil.exe` had already exited (expected for short `hashfile` runs), while `winpmem.exe` appears under the interactive Administrator session used for live acquisition (PID 2164; console-launched imager parentage, not a service or Office parent). `windows.malfind` returned multiple `PAGE_EXECUTE_READWRITE` regions: several under `MsMpEng.exe` (Microsoft Defender, PID 2488), plus additional RWX hits in `ServerManager.exe`, `wmiprvhost.exe`, and `wsmprovhost.exe`. Per the module lecture, Defender RWX regions are a known false-positive class and do *not* by themselves demonstrate attacker process injection. The non-Defender RWX hits are also consistent with legitimate Windows management/remoting components on a freshly built server and likewise do not prove staged LOLBin or Run-key activity injected code. Local `certutil` execution and the HKCU Run key leave no injected-VAD signature once those processes exit, so a malfind view dominated by Defender and built-in management hosts is the expected outcome for this staging and must not be over-read as “no attacker activity” or as confirmed injection.

3. Part B: Disk artifact triage

Evidence #4 – Prefetch (PECmd / CERTUTIL)

Evidence #4. CERTUTIL.EXE Prefetch row(s): run count, last-run timestamp(s), directories/files referenced.

PECmd against C:\Windows\Prefetch on MERIDIAN-APP01 (Windows Server 2022 Desktop Experience).

Prefetch configuration at collection time:

HKLM...\PrefetchParameters EnablePrefetcher=3 EnableSuperfetch=3
SysMain service = Running (Automatic)

PECmd version 2026.5.0

Found 0 Prefetch files

Processed 0 out of 0 files

CERTUTIL.EXE Prefetch row(s): NONE (0 .pf files after interactive + console staging).

Staging corroboration (pf_stage2.log, wall-clock 2026-10-02 20:18:11 -04:00):

certutil -hashfile C:\Windows\System32\calc.exe SHA256 (x6)

certutil -hashfile C:\Windows\System32\notepad.exe SHA256 (x1)

Each: "CertUtil: -hashfile command completed successfully."

SHA256(calc)=4208893c871d2499f184e3f0f2554da89f451fa9e98d95fc9516c5ae8f2b3bbd

SHA256(notepad)=b65079972e88691fe19b5d4d5eb3159f6cd627fb6c4f09ae9b9df959330082df

Server SKU did not materialize application Prefetch despite EnablePrefetcher=3, SysMain, ACL repair (SYSTEM FullControl), ProcessIdleTasks, and reboot.

Evidence #5 – Amcache

Evidence #5. Amcache entry for `certutil.exe` (or `calc.exe/notepad.exe`): SHA-1, path, first-execution timestamp.

AmcacheParser 2026.5.0 vs VSS Amcache.hve (+LOG1 replay) from MERIDIAN-APP01.

Command: AmcacheParser.exe -f Amcache.hve -i --csv <out>

New-format hive; 98 file entries after log replay.

Target search (certutil.exe / calc.exe / notepad.exe / win32calc.exe):

NO InventoryApplicationFile rows for those names.

Hive UTF-16 scan only hits VBoxCertUtil.exe (Guest Additions), not System32\certutil.exe.

Amcache.hve LastWrite stayed 2026-10-02 20:21:08 -04:00 after further staging.

Amcache OS-component rows that DID index (-i):

Name,FullPath,SHA1,FileKeyLastWriteTimestamp,IsOsComponent

powershell.exe,c:\windows\system32\windowspowershell\v1.0\powershell.exe,

5b1a36a8b206fe9daece7e2cd702cdd4e10bb895,2026-10-02 23:36:45,True

CompatTelRunner.exe,c:\windows\system32\compattelrunner.exe,

```
33999d1f1943f1319f2cac5b653966f0eecf7654,2026-10-02 23:36:20,True
```

```
Amcache ShortCuts (notepad.lnk, not notepad.exe execution):  
notepad.lnk, C:\ProgramData\...\Accessories\Notepad.lnk, 2026-10-02 23:37:42
```

```
On-host SHA-1 (Get-FileHash) of staged binaries:  
certutil.exe|C:\Windows\System32\certutil.exe|0d1f72efaf905a80fc5f1edc4e85889183f94fab  
calc.exe|C:\Windows\System32\calc.exe|80865a5563eb92b8653a5af3db371d7bc1674b2d  
notepad.exe|C:\Windows\System32\notepad.exe|58bf532f93949cc62821273f778b87da85787afa
```

```
AppCompatCache corroboration (exported AppCompatCache value):  
Path = C:\Windows\system32\certutil.exe
```

Evidence #6 – Registry Run key

Evidence #6. reg query output showing MeridianSyncHelper.

```
reg query "HKCU\Software\Microsoft\Windows\CurrentVersion\Run"
```

```
HKKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run  
MeridianSyncHelper REG_SZ powershell.exe -NoProfile -WindowStyle Hidden -Command "Get-Date | Out-File C:\  
Meridian\synclog.txt -Append"
```

```
2026-10-02 20:32:15 -04:00
```

Critical thinking #2

Prefetch supplied *no* last-run timestamp or run count on this Server 2022 host (Evidence #4: PECmd found 0 .pf files), so it cannot be used as a positive clock for `certutil`. Amcache FileKeyLastWriteTimestamp values that did index—`powershell.exe` at 2026-10-02 23:36:45 UTC and `CompatTelRunner.exe` at 23:36:20 UTC—sit in the same calendar-evening window as first-boot inventory, while the Section 1.2 wall-clock (2026-10-02 20:32:15 -04:00) and the `pf_stage2.log` `certutil` transcript (20:18:11 -04:00) mark the operator staging session about an hour later. AppCompatCache independently records `C:\Windows\system32\certutil.exe`, and Evidence #6 shows `MeridianSyncHelper` at that same wall-clock. Together these clocks corroborate lab-driven activity the same evening; the missing Prefetch/Amcache LOLBin rows are a Server SKU telemetry gap, not evidence that staging failed. Prefetch run count, had it existed, would still prove only that `certutil.exe` launched *N* times from an associated path—not which arguments were used or that any malicious action occurred.

4. Part C: Static malware-triage survey

Evidence #7 – static triage of `svcupdate.bin`

Evidence #7. Output of `file`, `sha256sum`, `strings -a -n 6`, and `xxd | head -n 12`.

```
=== file svcupdate.bin ===  
svcupdate.bin: data (minimal MZ stub; not a valid PE)  
  
=== sha256sum svcupdate.bin ===  
3f77970a8c10f1af034e984b3b06f76db1708d377e8c3f9085af95823c9e2d73  svcupdate.bin  
  
=== strings -a -n 6 svcupdate.bin ===  
MeridianLoader/2.1 (c) internal build  
MRD-STAGE2-INIT  
sync-update.meridian-relay.test  
  
=== xxd svcupdate.bin | head -n 12 ===  
00000000: 4d 5a 90 90 90 90 90 90 90 90 90 90 90 90 90 90  MZ.....  
00000010: 90 90 90 90 90 90 90 90 90 90 90 90 90 90 90 90  .....  
00000020: 90 90 90 90 90 90 90 90 90 90 90 90 90 90 90 90  .....  
00000030: 90 90 90 90 90 90 90 90 90 90 90 90 4d 65 72 69  .....Meri  
00000040: 64 69 61 6e 4c 6f 61 64 65 72 2f 32 2e 31 20 28  dianLoader/2.1 (
```

```

00000050: 63 29 20 69 6e 74 65 72 6e 61 6c 20 62 75 69 6c c) internal buil
00000060: 64 00 4d 00 52 00 44 00 2d 00 53 00 54 00 41 00 d.M.R.D.-.S.T.A.
00000070: 47 00 45 00 32 00 2d 00 49 00 4e 00 49 00 54 00 G.E.2.-.I.N.I.T.
00000080: 90 90 90 31 c0 50 68 00 00 00 00 73 79 6e 63 2d ...1.Ph....sync-
00000090: 75 70 64 61 74 65 2e 6d 65 72 69 64 69 61 6e 2d update.meridian-
000000a0: 72 65 6c 61 79 2e 74 65 73 74 00 00 00 00 00 00 relay.test.....
000000b0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Taxonomy classification

Classification: loader / downloader.

Three static indicators:

1. Banner string `MeridianLoader/2.1 (c) internal build` names the component as a loader family rather than ransomware or a worm.
2. UTF-16LE marker `MRD-STAGE2-INIT` is a stage-2 handoff token typical of multi-stage loaders that initialize a subsequent payload.
3. Embedded C2-style hostname `sync-update.meridian-relay.test` is a retrieval/beacon endpoint indicator aligned with downloader behavior, not encryption or kernel rootkit structures.

The file is only a minimal MZ stub (not a valid PE), matching the lab’s inert stand-in: structural taxonomy signals without runnable code.

Critical thinking #3

1. Weakest sole classification evidence: the leading MZ bytes. Benign Windows binaries also start with MZ, so that marker alone does not support malware taxonomy.
2. Strongest loader/downloader-specific indicator: the combination of the `MeridianLoader/2.1` banner with `MRD-STAGE2-INIT`, which encodes staged loading rather than ransomware notes, worm replication, or backdoor implant language by itself. The C2 hostname further supports remote retrieval over other categories.

5. Graded challenge: hunt hypothesis

Claim. After Parts A–C, MERIDIAN-APP01 shows operator-visible HKCU Run-key persistence (`MeridianSyncHelper`) and LOLBin execution evidence, but memory triage did not show injected code. A second, quieter persistence mechanism that would survive interactive logoff and avoid the Run key is a user or system Scheduled Task whose action launches `powershell.exe` or a lookalike helper under a benign task name.

Next artifact. Enumerate scheduled task XML under `C:\Windows\System32\Tasks` and the Task Scheduler registry keys (`HKLM\SOFTWARE\Microsoft\WindowsNT\CurrentVersion\Schedule\TaskCache`), searching for task actions that reference `powershell.exe`, `C:\Meridian`, or `sync-update.meridian-relay.test`. This fits the evidence because the staged Run key already abuses PowerShell for silent upkeep, and task XML is a parallel ATT&CK persistence class that would not appear in the HKCU Run query we already ran, nor necessarily in a clean `malfind`.

Negative result. If no such task exists, the investigation still advances: the Run key remains the only confirmed persistence on this host from Parts A–B, narrowing follow-up to service `ImagePath` values and WMI subscriptions rather than re-checking Run keys. Either outcome converts a vague “quieter persistence” concern into a tested, closed question grounded in disk artifacts memory alone cannot settle after process exit.

6. References

- NIST SP 800-61 Revision 2; IETF RFC 3227.
- Volatility 3 Documentation. <https://volatility3.readthedocs.io/>
- Velocidex WinPmem. <https://github.com/Velocidex/WinPmem>
- Eric Zimmerman / Kroll EZ Tools (PECmd, AmcacheParser). <https://ericzimmerman.github.io/>
- MITRE ATT&CK T1547.001 (Registry Run Keys); T1070.006 (Timestamp).